# BOXIFY CUSTOMER CHURN ANALYSIS

## Business Analytics Case Study

*A Comprehensive Data Analysis Project*

*For Junior Data Analysts*

Prepared by: Junior Data Analyst Team

Department: Business Intelligence & Analytics

## 1. Executive Summary

This report presents a comprehensive customer churn analysis for Boxify, an online subscription box service. The objective of the analysis was to identify the factors contributing to customer churn, evaluate customer lifetime value (CLV), analyze customer engagement patterns, and recommend strategies to improve customer retention.

The analysis revealed an overall churn rate of 18%, indicating that nearly one in five customers discontinued their subscription service. Monthly subscribers represented the largest proportion of churned customers, suggesting lower commitment levels compared to quarterly and annual subscription plans.

Customer engagement emerged as a key factor influencing retention. Active customers demonstrated higher engagement scores and significantly greater customer lifetime value than churned customers. The company also experienced approximately $1 million in revenue loss due to customer churn.

Customer segmentation analysis showed that churn varies across age groups, subscription plans, locations, and payment methods. Customers aged 36–45 exhibited the highest churn levels, while annual subscribers demonstrated stronger retention compared to monthly subscribers.

Based on these findings, the report recommends implementing targeted retention campaigns, improving customer engagement initiatives, promoting long-term subscription plans, and developing proactive churn prevention strategies. These initiatives can help reduce churn by 5–10%, improve customer loyalty, and increase long-term profitability.

Overall, the analysis highlights the importance of customer engagement and personalized retention strategies in sustaining business growth and maximizing customer lifetime value.

## 2. Business Scenario & Objectives

### 2.1 Company Profile

| Parameter | Details |
|---|---|
| Company Name | Boxify |
| Industry | E-commerce Subscription Services |
| Business Model | Subscription-Based |
| Products Offered | Curated Lifestyle Subscription Boxes |
| Subscription Plans | Monthly, Quarterly, Annual |
| Target Customers | Lifestyle and Convenience-Oriented Consumers |
| Analysis Period | Last 3 Years |
| Business Focus | Customer Retention and Loyalty Improvement |

### 2.2 Problem Statement

Over the past year, Boxify has experienced a significant increase in customer churn, with many subscribers canceling their subscriptions before completing six months. This trend has resulted in reduced customer lifetime value, revenue loss, and increased customer acquisition costs.

Management requires a data-driven analysis to identify the key factors contributing to churn, determine which customer segments are at the highest risk, and develop effective strategies to improve customer retention and loyalty.

### 2.3 Strategic Goals for Next Quarter

| Strategic Goal | Expected Outcome |
|---|---|
| Reduce customer churn by 5–10% | Improve customer retention |
| Increase customer engagement | Reduce cancellation risk |
| Identify high-risk customer segments | Enable targeted retention campaigns |
| Increase Customer Lifetime Value (CLV) | Improve long-term revenue |
| Promote long-term subscription plans | Strengthen customer loyalty |

## 3. Key Performance Indicators (KPIs)

To evaluate customer retention performance and understand the factors contributing to churn, several key performance indicators (KPIs) were analyzed. These metrics provide insights into customer behavior, engagement levels, revenue impact, and overall business performance.

| KPI | Description | Formula | Target |
| --- | --- | --- | --- |
| Churn Rate (%) | Percentage of customers who cancel their subscription | (Churned Customers ÷ Total Customers) × 100 | Reduce by 5–10% |
| Customer Lifetime Value (CLV) | Average revenue generated per customer during their subscription period | Total Revenue ÷ Total Customers | Increase CLV |
| Average Engagement Score | Average customer engagement level based on interactions | Total Engagement Score ÷ Total Customers | Increase Engagement |
| Average Subscription Duration | Average number of months customers remain subscribed | Total Subscription Months ÷ Total Customers | Increase Duration |
| Revenue Lost Due to Churn | Revenue associated with churned customers | Sum of TotalSpent for Churned Customers | Reduce Revenue Loss |
| Retention by Segment | Customer retention across age groups, locations, and subscription types | (Retained Customers ÷ Total Customers) × 100 | Improve Retention |

## 4. Dataset Overview

### 4.1 Data Specification

The dataset contains customer subscription information, engagement metrics, and spending behavior used to analyze customer churn patterns and retention performance.

| Column Name | Description |
|---|---|
| CustomerID | Unique identifier assigned to each customer |
| JoinDate | Date when the customer started the subscription |
| CancelDate | Date when the customer canceled the subscription (if applicable) |
| Age | Age of the customer |
| Gender | Gender of the customer |
| Location | Customer's city or geographic location |
| SubscriptionType | Type of subscription plan (Monthly, Quarterly, Annual) |
| TotalSpent | Total amount spent by the customer during the subscription period |
| BoxesReceived | Total number of subscription boxes received by the customer |
| EngagementScore | Customer engagement level based on interactions and activity |
| PaymentMode | Payment method used by the customer |
| ChurnStatus | Indicates whether the customer is Active or Churned (derived column) |
| AgeGroup | Customer age category used for segmentation analysis (derived column) |
| SubscriptionMonths | Total subscription duration in months (derived column) |

### 4.2 Data Preparation

To prepare the dataset for analysis, several transformations were performed to improve usability and support KPI calculations. Date fields were converted into the appropriate format, customer age groups were created for segmentation, churn status was derived based on cancellation information, and subscription duration was calculated in months. These transformations enabled accurate customer retention analysis and business insights.

## 5. Business Questions & Analytical Approach

**Q1: Which customer segments have the highest churn?**

**Approach**:

Customer churn was analyzed across key demographic and subscription-based segments, including age groups, geographic locations, and subscription types. Segmentation analysis was performed to identify customer groups with the highest cancellation rates and understand the characteristics associated with increased churn risk. The findings help support targeted retention strategies and more effective customer engagement initiatives.

**Q2: How does customer engagement influence churn?**

**Approach:**
Compare average engagement scores between active and churned customers. Visualize the results using column charts to identify the relationship between engagement and retention.

**Expected Findings**

- Active customers show higher engagement levels.
- Low engagement customers are more likely to churn.
- Engagement is a strong predictor of customer retention.

**Q3: What is the average lifetime value of churned vs active customers?**

**Approach:**
Customer Lifetime Value (CLV) was calculated and compared across active and churned customer groups. This comparison was used to assess the financial impact of customer churn on business performance.

| Customer Status | Expected Insight |
|---|---|
| Active Customers | Higher lifetime value and revenue contribution |
| Churned Customers | Lower lifetime value and reduced profitability |

**Q4: Can you predict potential churners based on historical data?**

**Approach:**
Historical customer behavior was analyzed to identify characteristics commonly associated with customer churn. Factors such as engagement level, subscription duration, and subscription type were examined to determine potential indicators of future churn risk.

| Historical Indicator | Churn Risk Interpretation |
|---|---|
| Low Engagement Score | Higher likelihood of customer churn |
| Short Subscription Duration | Indicates lower customer commitment |
| Monthly Subscription Plan | Associated with higher churn rates |
| Declining Customer Activity | Early sign of potential cancellation |
| Lower Customer Lifetime Value (CLV) | May indicate reduced long-term engagement |

**Q5: What retention strategies can be implemented to improve customer loyalty and reduce churn?**

**Approach:**
Insights from customer segmentation, engagement analysis, and churn behavior were used to develop targeted retention recommendations, including personalized engagement campaigns, loyalty programs, and subscription optimization strategies.

## 6. Exploratory Data Analysis (EDA) Guidance

### 6.1 Summary Statistics to Report

• Customer churn varies across age groups and subscription plans.

• Customer engagement has a strong influence on retention behavior.

• Long-term subscribers demonstrate better retention than monthly subscribers.

• Customer churn results in reduced customer lifetime value and revenue loss.

### 6.2 Recommended Visualizations

| Visualization | Business Question Answered |
|---|---|
| Churn by Age Group | Which age groups have the highest churn? |
| Churn by Subscription Type | Which subscription plans experience the highest churn? |
| Average Engagement by Churn Status | Does engagement influence customer retention? |
| Average CLV by Churn Status | How does customer value differ between active and churned customers? |
| Churn by Location | Which locations show higher churn levels? |
| Churn by Payment Mode | Which payment methods are associated with higher churn? |
| KPI Dashboard | What is the overall retention performance of the business? |

## 7. Expected Insights & Hidden Patterns

### 7.1: Engagement Impact on Churn

**Observation:**
Customers with lower engagement scores are more likely to cancel their subscriptions.

**Explanation:**
Low customer interaction indicates reduced interest in the service. Customers who engage less frequently with subscription content and offerings are more likely to discontinue their subscriptions, making engagement a key predictor of churn.

### 7.2: Subscription Commitment and Retention

**Observation:**
Monthly subscribers exhibit higher churn rates compared to quarterly and annual subscribers.

**Explanation:**
Customers on long-term subscription plans demonstrate stronger commitment and loyalty. Longer subscription periods reduce cancellation opportunities and encourage continued engagement with the service.

### 7.3: Customer Lifetime Value and Churn

**Observation:**
Active customers generate significantly higher Customer Lifetime Value (CLV) than churned customers.

**Explanation:**
Customers who remain subscribed for longer periods contribute more revenue and receive more subscription boxes. Reducing churn can therefore increase customer lifetime value and improve overall business profitability.

## 8. Recommendations

### 8.1: Quick Wins to Reduce Churn

| Recommendation | Expected Impact |
|---|---|
| Personalized email campaigns | Improve engagement |
| Renewal discounts | Reduce cancellations |
| Special offers for inactive users | Increase retention |

### 8.2: Long-Term Strategies to Reduce Churn

**Customer Loyalty Program**

- Reward long-term subscribers through points and exclusive benefits.
- Helps improve customer loyalty and retention.

**Subscription Plan Optimization**

- Encourage migration from monthly to annual plans.
- Increases customer commitment and reduces churn.

**Predictive Churn Monitoring**

- Identify at-risk customers before cancellation.
- Enables proactive retention efforts.

### 8.3: Engagement Strategies for High-Risk Customers

| High-Risk Customer Group | Engagement Strategy |
|---|---|
| Low Engagement Customers | Personalized reminders and exclusive offers |
| Monthly Subscribers | Upgrade incentives for long-term plans |
| New Customers | Onboarding campaigns |
| Inactive Customers | Re-engagement campaigns |

## 9. Constraints & Assumptions

| Constraint / Assumption | Implication for Analysis |
|---|---|
| Data is available only for the last 3 years. | Analysis is limited to historical trends within the available period. |
| Monthly subscription renewals are assumed. | Customer retention and churn calculations are based on monthly renewal cycles. |
| Churn can only be measured at the end of a subscription cycle. | Early churn indicators may not be fully captured in the analysis. |
| No external customer behavioral data is included. | Insights are derived solely from the available dataset and may not reflect external influences on customer behavior. |

## 10. Deliverables & Submission Checklist

| Deliverable | Description |
|---|---|
| Power BI Dashboard | Interactive dashboard showing churn KPIs and customer retention insights |
| Customer Churn Analysis Report | Comprehensive report covering analysis, insights, and recommendations |
| KPI Analysis | Evaluation of churn rate, CLV, engagement, subscription duration, and revenue loss |
| Customer Segmentation Analysis | Analysis of churn across age groups, locations, and subscription types |
| Business Recommendations | Data-driven strategies to improve customer retention and reduce churn |
| Final PDF Submission | Consolidated report prepared for submission and review |

### 10.1 Report Evaluation Criteria

| Criteria | Weightage |
|---|---|
| Business Understanding & Problem Definition | 15% |
| Data Preparation & Transformation | 15% |
| Exploratory Data Analysis (EDA) | 15% |
| Dashboard Design & Visualization | 20% |
| Business Insights & Interpretation | 20% |
| Recommendations & Strategic Thinking | 10% |
| Report Presentation & Documentation | 5% |